

1. Conditional Grid Walker

Problem:

You are given an $n \times m$ grid. Each cell contains an integer. You start at (1,1) and want to reach (n,m).

Rules:

- You can move right or down.
- If the current value is:
 - Even $\rightarrow$ you can move normally
 - Odd $\rightarrow$ you must skip one cell (jump 2 steps)
 - Negative $\rightarrow$ you must go back one step (if possible), else stop

Find the number of ways to reach (n,m).

Input

t
n m
grid...

Output

Number of ways for each test case

2. Multi-Layered FizzBuzz Variant

Modify FizzBuzz:

- If divisible by 3 $\rightarrow$ "Fizz"
 - If divisible by 5 $\rightarrow$ "Buzz"
 - If divisible by 7 $\rightarrow$ reverse digits
 - If prime $\rightarrow$ print square
 - If divisible by both 3 and 5 $\rightarrow$ skip output
 - Stop early if number contains digit 0
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3. Nested Loop Simulation Game

Simulate a game:

- Player energy starts at E
- n rounds
- Each round:
 - If energy > threshold $\rightarrow$ attack
 - Else $\rightarrow$ defend
 - Special rounds: multiples of 4, 7, 11 change rules

Track final energy and actions taken.

4. Pattern Matrix with Conditional Fill

Fill an $n \times n$ matrix:

- Spiral order
 - If index sum is even $\rightarrow$ fill increasing numbers
 - Else $\rightarrow$ fill decreasing numbers
 - Skip multiples of 5
 - Replace primes with 0
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5. Dynamic Sequence Validator

Given an array:

- If increasing $\rightarrow$ continue
 - If decreasing $\rightarrow$ reverse subarray
 - If equal $\rightarrow$ skip
 - If violates both $\rightarrow$ break and output index
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6. Multi-Test Conditional Sorting

Sort array with rules:

- If element is even $\rightarrow$ move left
 - If odd $\rightarrow$ move right
 - If divisible by 3 $\rightarrow$ stay fixed
 - If negative $\rightarrow$ reverse entire array
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7. Complex Number Transformation

Repeat operations until stable:

- If even $\rightarrow$ divide by 2
 - If odd $\rightarrow$ multiply by 3 + 1
 - If number $>$ limit $\rightarrow$ reset to initial
 - Stop if cycle detected or steps exceed k
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8. Grid Escape with Traps

Navigate grid:

- # $\rightarrow$ wall
- T $\rightarrow$ trap (skip next move)
- B $\rightarrow$ bonus (extra move)
- X $\rightarrow$ teleport

Find minimum steps or detect infinite loop.

9. Conditional String Processor

Given string:

- If vowel → uppercase
 - If consonant → lowercase
 - If digit → repeat digit times
 - If special char → remove previous char
 - Stop if length exceeds limit
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10. Multi-Level Loop Breaker

Process numbers:

- Outer loop over test cases
- Inner loop:
 - If divisible by 2 → continue
 - If divisible by 3 → break
 - If prime → store
 - If palindrome → reverse digits
- Print final transformed sequence